

IEAP-2026 Series 03 — Data wrangling and ANOVA

RStudio assignment · Team TUBA / DEEMUA / GILLES

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About this report

- **GitHub repository (public):** <https://github.com/ChloeGil-student/IEAP-Series03-RStudio>
- **Goal:** reproduce the main result of Mottet et al. (2017) (their Fig. 3: movement time as a function of the index of difficulty, by group) with a full ANOVA pipeline, and discuss why our numbers differ from the published ones.
- **Structure:** this file is the *master document*. Each part is written in its own file in **Sections/** and included here at render time. Section numbers 1.1 to 1.7 are those of the series.

Section	File	Owner
1.1–1.3 Question, data reorganization, ANOVA	Sections/01_Data-reorganisation.qmd	Chloé GILLES
1.4 Linear regression by GROUP	Sections/02_Linear-regression-by-group.qmd	Charles Sam DEEMUA
1.5–1.6 Graph and PDF export	Sections/03_Save-the-graph.qmd	Tuba TUBA
1.7 Discussion of the differences	Sections/04_Discussion-of-the-differences.qmd	Chloé GILLES
2–4 Git workflow, challenges, checklist	Sections/05_GitWorkflow.qmd, Sections/06_Checklist.qmd	All together

Setup

The whole report starts with one command. It runs `R/setup.R`, which loads the libraries, reads the data and builds the clean data set used by every section.

```
source(here::here("R", "setup.R"))

theme_set(theme_classic(base_size = 11)) # same look for all graphs
```

This is the content of `R/setup.R`. The reasons for the cleaning step are given in section 1.3.

```
# R/setup.R
# Shared setup: libraries + data loading + cleaning.
# Sourced once at the top of the master document.
library(ez)
library(tidyverse)
library(here)

# Load the raw data
Results <- read.table(here("data", "Results.txt"), header = TRUE, sep = " ",
                      stringsAsFactors = TRUE)

# Clean data: remove ID3 and REP3 (incomplete conditions)
clean_results <- Results %>%
```

```

filter(ID != "ID3") %>%
filter(REP != "R3") %>%
droplevels() %>%
mutate(
  GROUP = factor(GROUP, levels = c("pp", "ap", "cp")),
  ID_num = ifelse(ID == "ID1", 3.80, 5.43)
)

```

1 ANOVA: does the effect of difficulty depend on the group?

1.1 Think before coding

The article (Mottet et al. 2017) compares three groups on a reaching task performed at two levels of difficulty. Its Fig. 3 plots movement time (MT) against the index of difficulty (ID) for each group. Our scientific question is therefore:

Does the effect of task difficulty on movement time depend on the group?

Before writing any code, we fixed what each part of the question means in the data file.

In the question	In Results.txt	Role in the analysis
Movement time	MovementTime (s)	dependent variable
Difficulty	ID (ID1, ID2, ID3)	within -subject factor: a participant does several levels
Group	GROUP (pp, ap, cp)	between -subject factor: a participant belongs to one group
Participant	SUBJ	unit of observation

The article has 19 patients, 13 age-matched controls and 19 young controls. We will check in section 1.3 that the group sizes in the file match, which tells us that **pp** are the patients, **ap** the age-matched controls and **cp** the young controls.

The question then splits into the four sub-questions of the series:

1. *Is the effect of difficulty the same in all groups?* This is the **GROUP × ID interaction**, not the two main effects.
2. *How do we visualize it?* With one line per group across the ID levels: parallel lines mean no interaction.
3. *Is there a significant difference?* We test it with a mixed ANOVA (GROUP between, ID within).
4. *What sentence do we write in the report?* See the end of section 1.3.

One point needed attention from the start: the file has one row per reaching movement, but the ANOVA compares participants. Movements of the same person are not independent observations, so they must be averaged per participant before the test.

1.2 Which library?

We use `ez`, as suggested in the series: `ezANOVA()` runs a mixed ANOVA in one call. We looked at what it does with our file: when it receives several rows per participant and condition, it averages them itself and only says so in a warning. We prefer to do this averaging ourselves (section 1.3), so that the step is visible in the report and the ANOVA receives exactly one value per participant and ID level.

1.3 Data reorganization

The data are loaded by `R/setup.R` (see *Setup* at the start of the report). The raw table `Results` has 7560 rows, one per reaching movement.

1.3.1 Looking for missing data

A within-subject factor needs every participant in every level. As in the series, we first count the rows in each `GROUP × ID` cell.

```
Results %>%
  count(GROUP, ID) %>%
  pivot_wider(names_from = ID, values_from = n)
```

```
# A tibble: 3 x 4
  GROUP ID1 ID2 ID3
  <fct> <int> <int> <int>
1 ap      740   740   740
2 cp     1140  1140  1140
3 pp      760   760   400
```

The group `pp` has about half the expected rows at `ID3`. Counting rows shows that something is missing, but not who. Counting participants answers both questions at once, for `ID` and for the repetition `REP`:

```
Results %>%
  group_by(GROUP, ID, REP) %>%
  summarise(participants = n_distinct(SUBJ), .groups = "drop") %>%
  pivot_wider(names_from = REP, values_from = participants)
```

```
# A tibble: 9 x 5
  GROUP ID    R1  R2  R3
  <fct> <fct> <int> <int> <int>
1 ap    ID1     13   13   11
2 ap    ID2     13   13   11
3 ap    ID3     13   13   11
4 cp    ID1     19   19   19
```

5	cp	ID2	19	19	19
6	cp	ID3	19	19	19
7	pp	ID1	19	19	NA
8	pp	ID2	19	19	NA
9	pp	ID3	10	10	NA

The group sizes are 13, 19 and 19, as in the article. Two conditions are incomplete:

- **ID3** was done by only 10 of the 19 patients. Keeping it would mean either losing 9 patients or comparing the groups on different conditions.
- **R3** was done by no patient and by 11 of the 13 age-matched controls.

1.3.2 Two decisions

We remove ID3 for everybody. The interaction cannot be estimated fairly with a level that half of one group did not do.

We also remove R3 for everybody. This one is a choice, not an obligation: the ANOVA would run with R3, because the rows are averaged per participant anyway. We had two reasons. The first is in the data: the first repetition is slower than the following ones.

```
Results %>%
  filter(ID != "ID3") %>%
  group_by(GROUP, REP) %>%
  summarise(mean_MT = round(mean(MovementTime), 2), .groups = "drop") %>%
  pivot_wider(names_from = REP, values_from = mean_MT)
```

```
# A tibble: 3 x 4
  GROUP    R1    R2    R3
  <fct> <dbl> <dbl> <dbl>
1 ap     1.17  1.1   1.13
2 cp     0.99  0.88  0.89
3 pp     4.22  3.55  NA
```

With R3, the average of a control participant would include an extra, practised repetition that no patient did, and part of the group difference would be a practice effect. The second reason is the design of the article: two levels of difficulty, and 80 reaching movements per participant (Mottet et al. 2017, Methods). Without ID3 and R3, our data have exactly this structure (checked below).

1.3.3 The cleaning rule

As in the series, we recount after cleaning. Every cell is now filled, and within a group all cells have the same number of rows:

Both decisions are written once, in the `filter()` line of `R/setup.R` (shown in *Setup*), so that every section works on the same `clean_results`.

```
clean_results %>%
  count(GROUP, ID, REP) %>%
  pivot_wider(names_from = REP, values_from = n)
```

```
# A tibble: 6 x 4
  GROUP ID      R1    R2
  <fct> <fct> <int> <int>
1 pp    ID1      380   380
2 pp    ID2      380   380
3 ap    ID1      260   260
4 ap    ID2      260   260
5 cp    ID1      380   380
6 cp    ID2      380   380
```

The cleaning keeps 4080 of the 7560 rows (54 %). The price is that we set aside valid data, mostly from the two control groups.

1.3.4 One value per participant and ID level

```
subject_means <- clean_results %>%
  group_by(GROUP, SUBJ, ID, ID_num) %>%
  summarise(n_movements = n(),
            MovementTime = mean(MovementTime),
            .groups = "drop")

# Safety check: every participant must have both ID levels
stopifnot(all(count(subject_means, SUBJ)$n == 2))
```

We check the result of the cleaning on this table: number of participants per group, and number of movements behind each participant mean.

```
subject_means %>%
  group_by(GROUP) %>%
  summarise(participants = n_distinct(SUBJ),
            min_movements = min(n_movements),
            max_movements = max(n_movements))
```

```
# A tibble: 3 x 4
  GROUP participants min_movements max_movements
  <fct>          <int>          <int>          <int>
1 pp              19             40             40
2 ap              13             40             40
3 cp              19             40             40
```

Every participant mean is based on 40 movements, so the design is balanced within participants, and 2×40 gives the 80 movements of the article. The groups, however, do not have the same number of participants: the design is not balanced between groups. Cleaning cannot fix that, and it is the reason for `type = 3` in the ANOVA below.

1.3.5 Visualize the interaction

We draw the interaction plot of the series: the mean movement time in each `GROUP × ID` cell, with error bars of half a standard deviation on each side, computed over all the movements of the cell.

```
ggplot(clean_results) +
  aes(x = ID, y = MovementTime, group = GROUP,
      color = fct_relevel(GROUP, "ap", "cp")) + # group order of the series
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun = mean, geom = "errorbar", width = 0.01,
              fun.min = \(x) mean(x) - sd(x) / 2,
              fun.max = \(x) mean(x) + sd(x) / 2) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size = 3) +
  labs(title = "Interaction between GROUP and ID on MovementTime",
       x = "ID", y = "Mean MovementTime (s)", color = "GROUP") +
  theme_minimal()
```

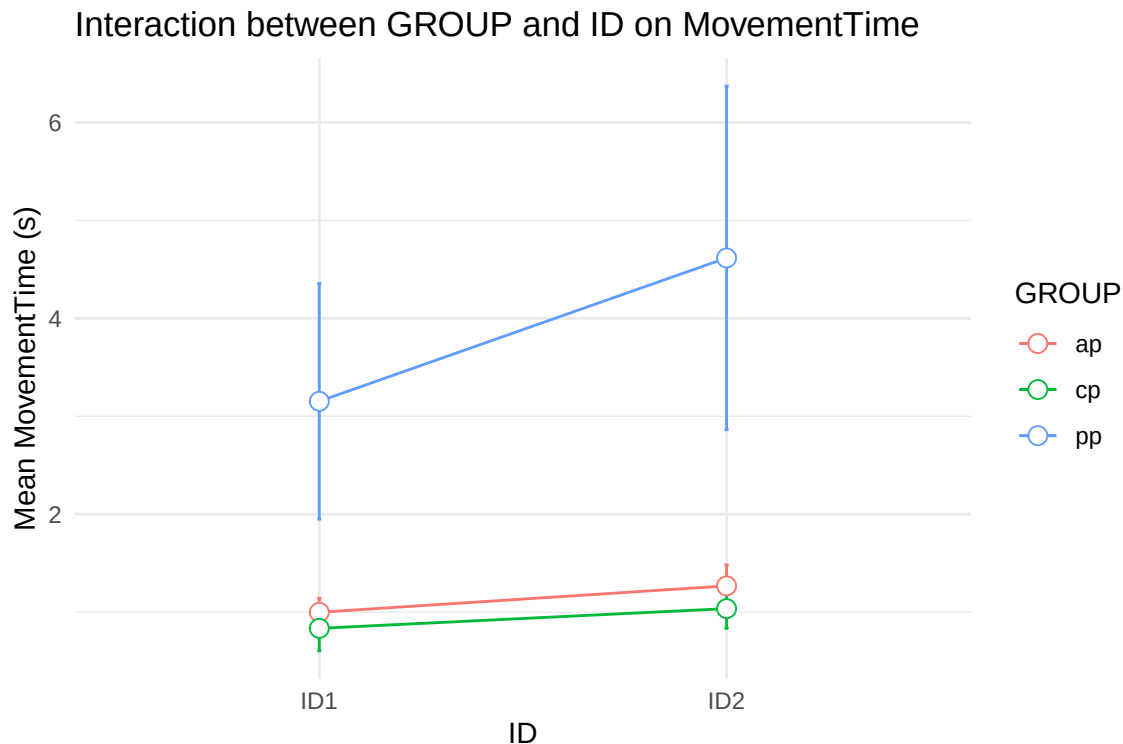


Figure 1: Interaction between GROUP and ID. Points: mean of all movements of the cell; bars: plus or minus half a standard deviation of these movements.

The lines are not parallel in Figure 1: movement time rises much more between ID1 and ID2 for the patients than for the two control groups.

The error bars of this plot describe the spread of individual movements. The ANOVA, however, works on one value per participant, so what matters for the test is the spread between participants. The table below gives the mean and the standard deviation of the participant means in each cell. The means are the same as in the plot, because every participant has the same number of movements; the standard deviations are not.

```
cell_means <- subject_means %>%
  group_by(GROUP, ID, ID_num) %>%
  summarise(n = n(),
            mean = mean(MovementTime),
            sd = sd(MovementTime),
            .groups = "drop")

knitr::kable(cell_means, digits = 2)
```

GROUP	ID	ID_num	n	mean	sd
pp	ID1	3.80	19	3.15	1.47
pp	ID2	5.43	19	4.62	2.41
ap	ID1	3.80	13	1.00	0.20
ap	ID2	5.43	13	1.27	0.29
cp	ID1	3.80	19	0.83	0.12
cp	ID2	5.43	19	1.04	0.11

This table shows a problem for the ANOVA, which assumes similar variances in all groups: the largest standard deviation is 22 times the smallest. We come back to this in section 1.7.

1.3.6 Run the ANOVA

```
res_anova <- ezANOVA(
  data = as.data.frame(subject_means), # ez expects a plain data frame
  dv = MovementTime,
  wid = SUBJ,
  between = GROUP,
  within = ID,
  type = 3 # groups of unequal size
)
```

Warning: Data is unbalanced (unequal N per group). Make sure you specified a well-considered value for the type argument to ezANOVA().

```
res_anova$ANOVA
```

	Effect	DFn	DFd		F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*		0.57462110
3	ID	1	48	26.10886	5.543011e-06	*		0.06581307
4	GROUP:ID	2	48	11.58124	7.866206e-05	*		0.05882293

Warnings are hidden everywhere else in this report, but not for this chunk: `ezANOVA()` reminds us that the groups are of unequal size, which is what we saw above. There is no sphericity test in the output because ID has only two levels. The table is identical to the one in the series, which confirms that averaging per participant ourselves is what `ezANOVA()` was doing silently.

1.3.7 The sentence for the report

Movement time differed between the groups (main effect of GROUP, $F(2, 48) = 37.24$, $p < .001$) and was longer for the more difficult task (main effect of ID, $F(1, 48) = 26.11$, $p < .001$). **The effect of difficulty was not the same in all groups** (GROUP $\times$ ID interaction, $F(2, 48) = 11.58$, $p < .001$): the increase of movement time with difficulty was steeper for the patients (Figure 1).

The effect sizes put this result in perspective. The generalized eta squared (column `ges`) is 0.57 for GROUP but only 0.06 for the interaction. The main difference between the groups is that patients are slower at *both* levels of difficulty; the fact that difficulty costs them more is real, but it comes second.

The ANOVA says that the groups differ, not which ones. We did not run post-hoc comparisons, so the statement that the *patients* differ from the two control groups rests on the figure, not on a test.

1.4 Linear regression by GROUP

The ANOVA treats ID1 and ID2 as two labels. The article goes one step further and describes each group by a line, $MT = a + b \times ID$ (Fitts's law), where ID is a number of bits, the slope b is the time cost of one more bit of difficulty and the intercept a is the predicted time for a movement without accuracy constraint (Mottet et al. 2017). We compute this line for each group.

1.4.1 ID as a number

A regression needs a numerical predictor. In the article, the two levels of difficulty are 3.80 and 5.43 bits (Mottet et al. 2017, Methods). The conversion is done once, in `R/setup.R` (column `ID_num`, shown in *Setup*). We check that each label received the right value:

```
clean_results %>%  
  count(ID, ID_num)
```

	ID	ID_num	n
1	ID1	3.80	2040
2	ID2	5.43	2040

1.4.2 One regression per group

The function below fits the line for each group of a data frame and returns the intercept, the slope and the equation as text. We wrote a function because we fit the same model on other versions of the data in section 1.7.

```
fit_by_group <- function(data) {
  data %>%
    group_by(GROUP) %>%
    group_modify(~ {
      model <- lm(MovementTime ~ ID_num, data = .x)
      tibble(intercept = coef(model)[[1]], slope = coef(model)[[2]])
    }) %>%
    ungroup() %>%
    mutate(equation = sprintf("MT = %.3f + %.3f * ID", intercept, slope))
}

regression_by_group <- fit_by_group(clean_results)

knitr::kable(regression_by_group, digits = 3)
```

GROUP	intercept	slope	equation
pp	-0.259	0.898	MT = -0.259 + 0.898 * ID
ap	0.372	0.165	MT = 0.372 + 0.165 * ID
cp	0.366	0.123	MT = 0.366 + 0.123 * ID

These are the three equations given in the series, so our cleaning and our conversion of ID reproduce the expected result.

1.4.3 What these equations do and do not tell us

What they tell us. The slope is positive in the three groups: movement time increases with difficulty. One more bit costs 0.9 s to the patients, against 0.17 s and 0.12 s to the age-matched and young controls, that is 5.4 and 7.3 times more. The difference between the slopes is the $\text{GROUP} \times \text{ID}$ interaction of section 1.3, expressed in seconds per bit.

What they do not tell us. We see three limits.

- *The regression adds no information to the two means.* With only two values of ID, the least-squares line of a group passes exactly through its two mean movement times: the slope is $(\text{mean at ID2} - \text{mean at ID1}) / (5.43 - 3.80)$. We cannot test whether the relation is really

linear, a limit that the article also states. The file does contain a third level, but not all patients did it and we do not know its value in bits.

- *The intercept is an extrapolation.* It is the predicted movement time at $ID = 0$, which is 3.8 bits away from the nearest measurement, so a small change in the two means is amplified. The negative intercept of the patients cannot be a movement time; it only says where the extended line crosses the vertical axis.
- *We report coefficients, not their uncertainty.* The model is fitted on all movements as if they were independent, although they come from 51 participants. This does not change the intercepts and slopes, but the standard errors and p-values that `lm()` would print are too optimistic, so we do not report them. Our test of the difference between groups is the ANOVA, which is based on participants.

1.5 Graph of movement time as a function of ID

The goal is a graph that resembles Fig. 3 (left panel) of the article: for each group, the mean movement time at the two levels of difficulty, error bars, the regression line and its equation.

Everything we need already exists: the means and standard deviations are in `cell_means` (section 1.3) and the lines and equations in `regression_by_group` (section 1.4). This section only draws.

1.5.1 Choices we made

- **Error bars.** The caption of Fig. 3 says that the bars are standard deviations between participants. We use the `sd` column of `cell_means`.
- **Lines.** The regression lines are drawn over the whole x axis, as in the article.
- **Axes.** The x axis starts at 0 so that the intercepts are visible, and the y axis goes below 0 because the line of the patients does.
- **Equations.** Their text comes from `regression_by_group`, so the graph cannot disagree with the table of section 1.4.

```
# Position of the three equations, one under the other
equation_labels <- regression_by_group %>%
  mutate(x = 0.3, y = 6.9 - 0.4 * row_number())

graph_MT_ID <- ggplot(cell_means) +
  aes(x = ID_num, y = mean, color = GROUP) +
  geom_abline(
    data = regression_by_group,
    aes(intercept = intercept, slope = slope, color = GROUP),
    linetype = "dashed",
    show.legend = FALSE
  ) +
  geom_errorbar(
    aes(ymin = mean - sd, ymax = mean + sd),
    width = 0
  ) +
  geom_point(
```

```

    shape = 21,
    fill = "white",
    size = 3
) +
geom_text(
  data = equation_labels,
  aes(x = x, y = y, label = equation),
  hjust = 0,
  show.legend = FALSE
) +
scale_x_continuous(breaks = 0:6) +
coord_cartesian(
  xlim = c(0, 6.2),
  ylim = c(-1, 7.5),
  expand = FALSE
) +
labs(
  x = "ID (bit)",
  y = "Movement time (s)"
)

graph_MT_ID

```

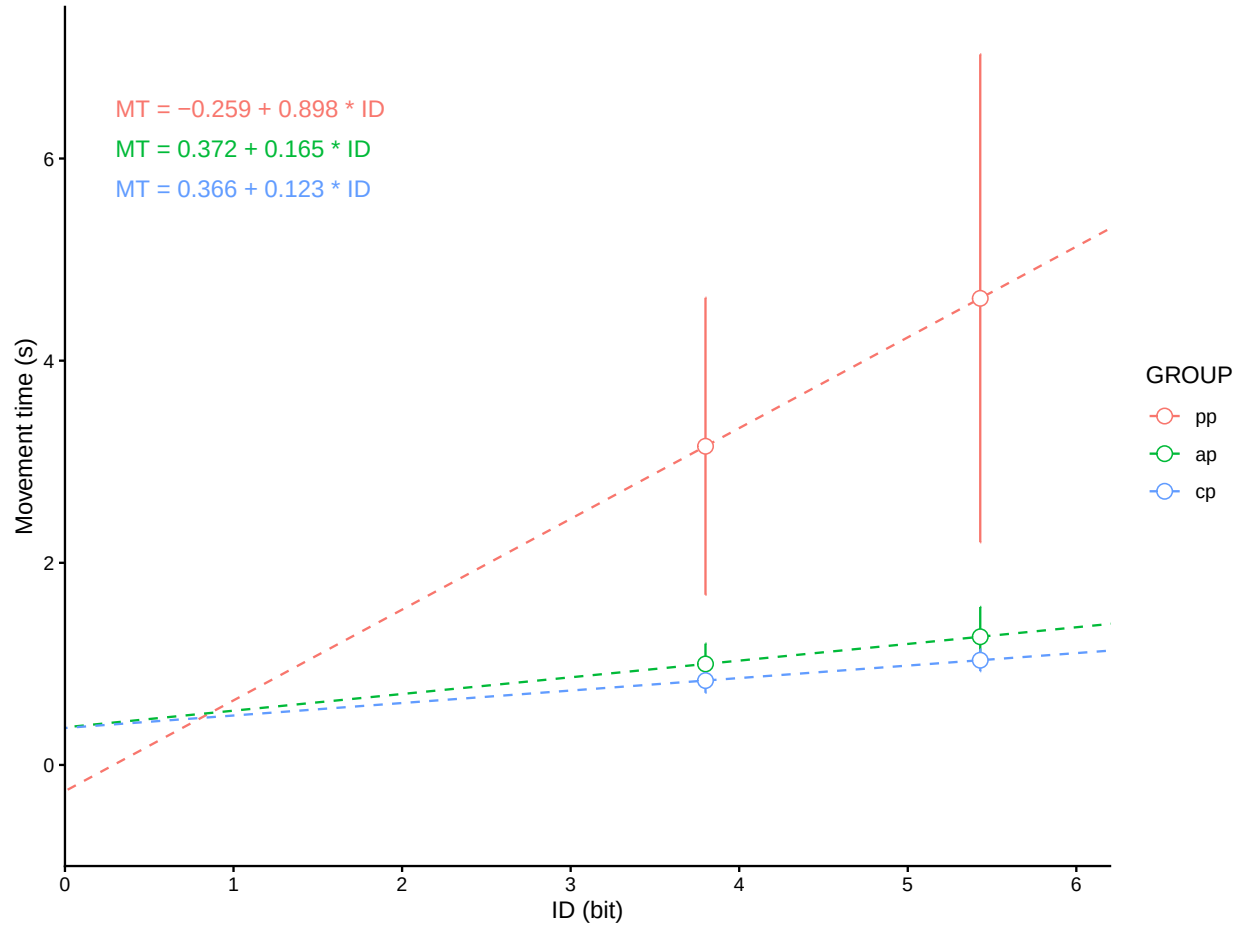


Figure 2: Movement time as a function of the index of difficulty. Points: group means; bars: one standard deviation between participants; dashed lines: linear regression of each group.

1.5.2 Reading the graph

Figure 2 has the same shape as Fig. 3 of the article: two almost flat and almost identical lines for the control groups, and a much higher and steeper line for the patients, with error bars that are several times longer. Two differences are visible without any calculation. Our points for the patients are higher, especially at the more difficult level, and our line for the patients crosses the vertical axis below zero, whereas the three lines of the article start close together around half a second. Section 1.7 looks for the reason.

1.6 Export the graph as a PDF

The journal asks for a figure of 8×6 inches. The file is written in the folder `figures/` of the repository.

```
dir.create(here("figures"), showWarnings = FALSE)
```

```
ggsave(here("figures", "MT_as_function_of_ID.pdf"),
       plot = graph_MT_ID, width = 8, height = 6, units = "in")
```

Data sources

File	Description	Source (URL)
data/Results.txt	One row per reaching movement: GROUP, SUBJ, TRIAL, ORI, DIR, ID, REP, NbVelPeaks, MovementTime	Course Moodle page, IEAP Series 03: https://moodle.umontpellier.fr/mod/folder/view.php?id=1025707
Fig. 3 of the article	Reference figure and equations (MT as a function of ID, by group)	https://doi.org/10.1371/journal.pone.0173674

Software

Analyses were run in R (R version 4.6.1 (2026-06-24)) with the packages `ez` (4.5.0), `tidyverse` (2.0.0) and `here` (1.0.2). The report was written with Quarto.

References

Mottet, Denis, Liesjet Elisabeth Henriette van Dokkum, Jérôme Froger, Abdelkader Gouaïch, and Isabelle Laffont. 2017. “Trajectory Formation Principles Are the Same After Mild or Moderate Stroke.” *PLOS ONE* 12 (3): e0173674. <https://doi.org/10.1371/journal.pone.0173674>.